Constructor University Bremen

MPFR LAB II

Part 3: Simulation of the Motor Model
Spring Semester 2025

Task1.10 - Task1.16

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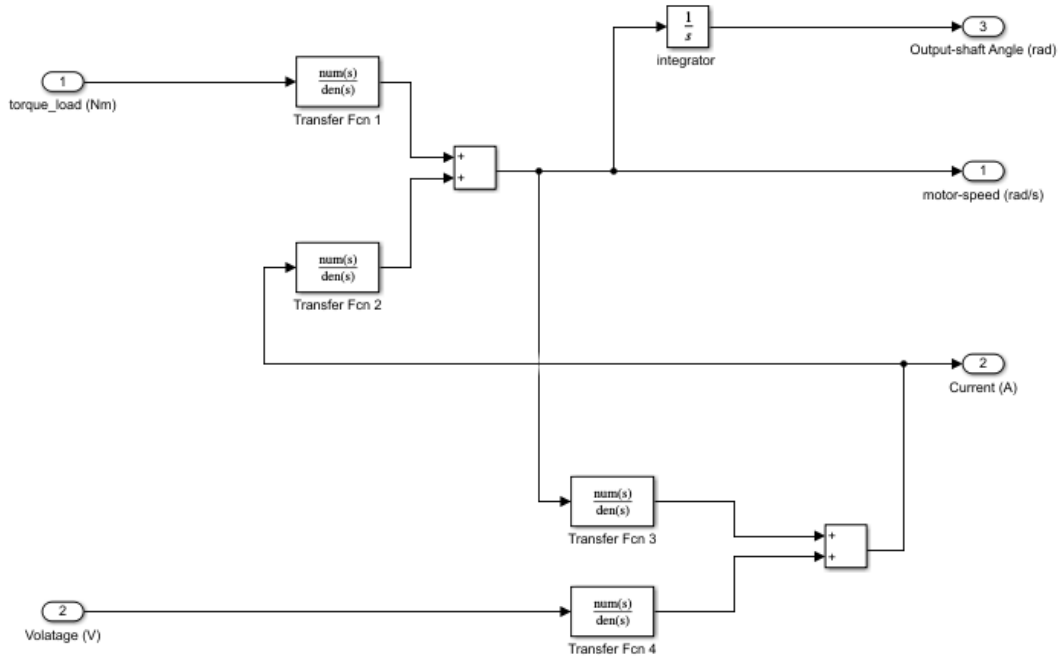
Task 1.10

1. *create the model as shown in Fig. 1.10. Verify that it is indeed implanting the equations (1.24a) and (1.24b).*



Transfer-function 1, $\frac{1}{Js+b}$

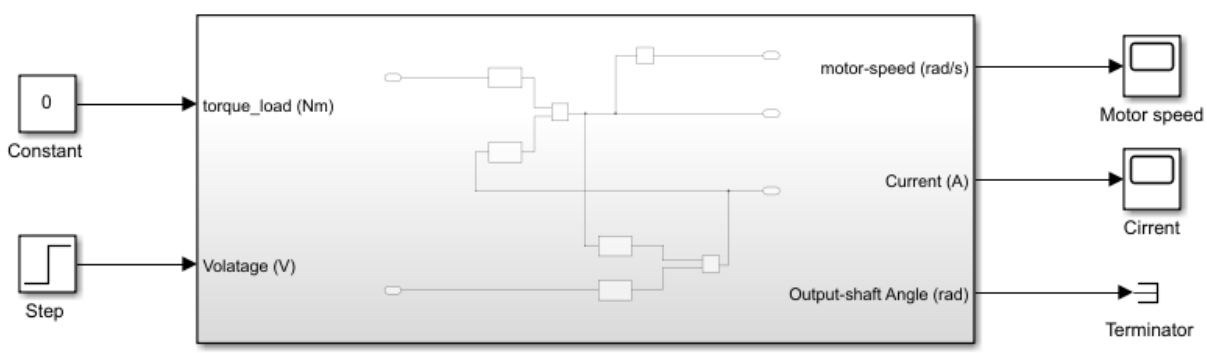
Transfer-function 2, $\frac{kt}{Js+b}$

Transfer-function 3, $\frac{ke}{Ls+R}$

Transfer-function 4, $\frac{1}{Ls+R}$

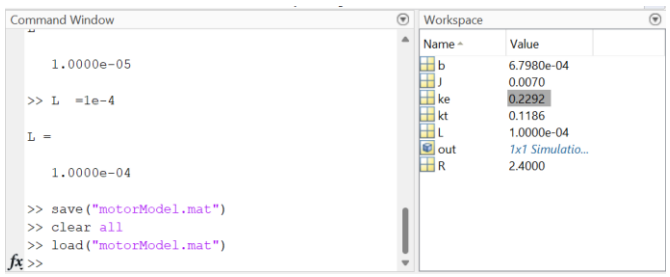
Workspace

Name	Value
b	6.7980e-04
J	0.0070
ke	0.2292
kt	0.1186
L	1.0000e-04
out	1x1 Simulatio...
R	2.4000

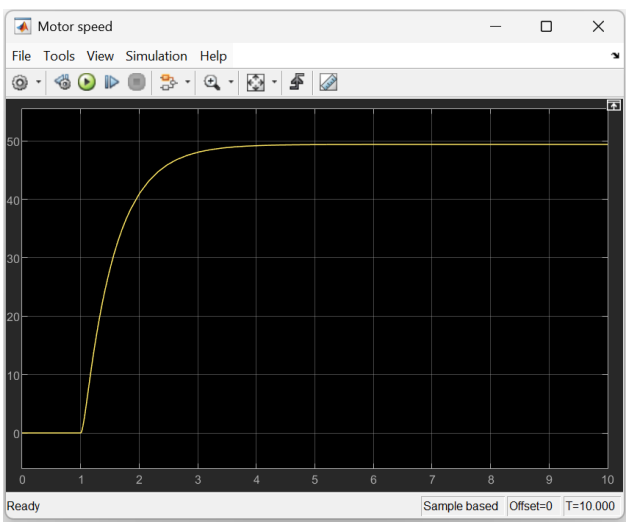


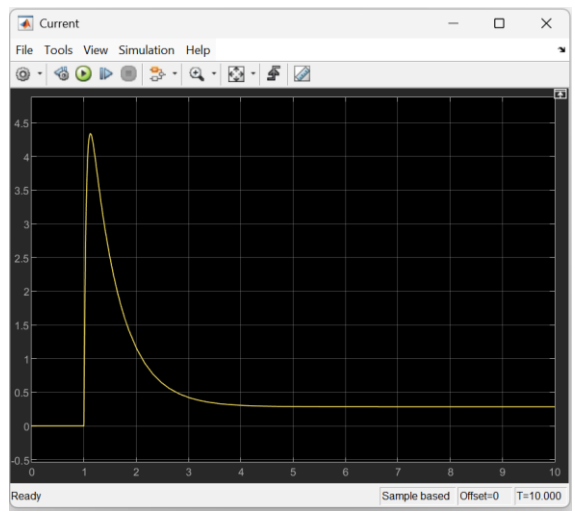
2. In the Matlab workspace create the variables representing the motor-parameters such as kt, ke, Ra, La, J, b, etc.

- *Type save('motor_params.mat') on the command line to save these values for later use.*
- *To make sure that these parameter values are automatically loaded when you open this model the next time, navigate to the menu: File/Model Properties/Model Properties, click on the "Callbacks" tab of the dialog-box that pops up, and under the "PreLoadFcn" callback, type load('motor_params.mat') on the right pane. Click OK and then save the model. You can test by closing the model, typing clear all to clear the workspace, and opening the model again.*

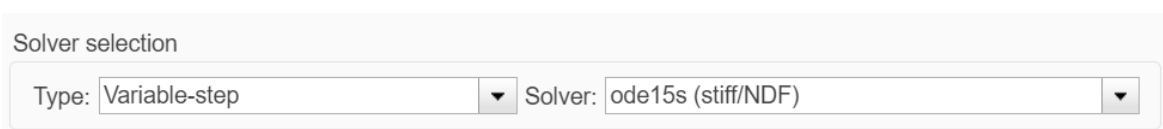


3. Apply a step of 12 V at t=1. Keep the load-torque to 0. Simulate the system from 0 to 10 seconds using the default solver (ode45).





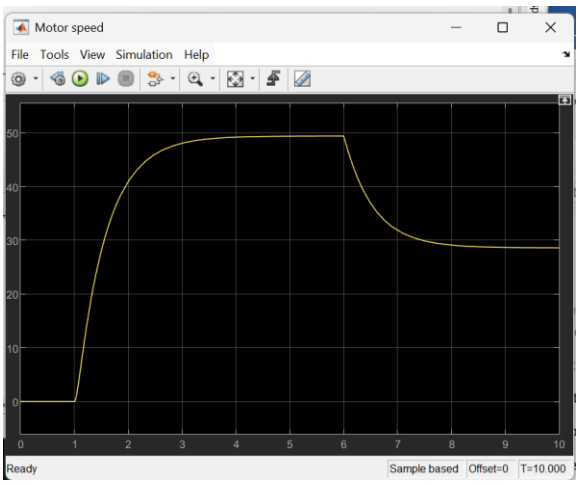
4. If the results shown in the scopes do not appear properly, press Ctrl+E and switch the solver to ode15s (stiff/NDF). Press the "Autoscale" button of the scope to see the whole curve.

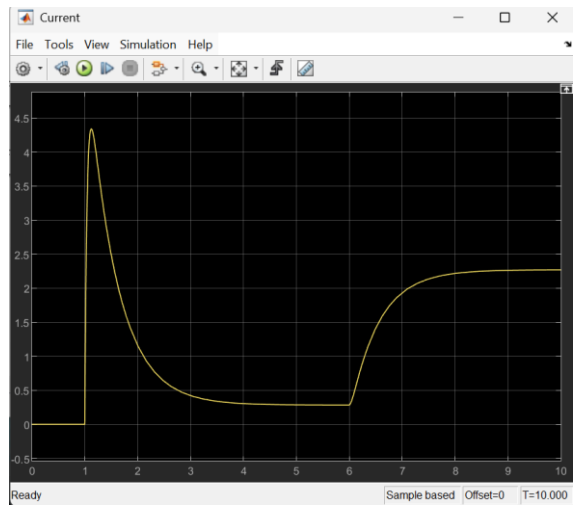


5. Explain the behavior of the current (the initial peak) and the speed scopes. Hint: Make use of (1.18). Verify that they settle to the no-load values in Table 1.1? You can use the zoom button in the scope.

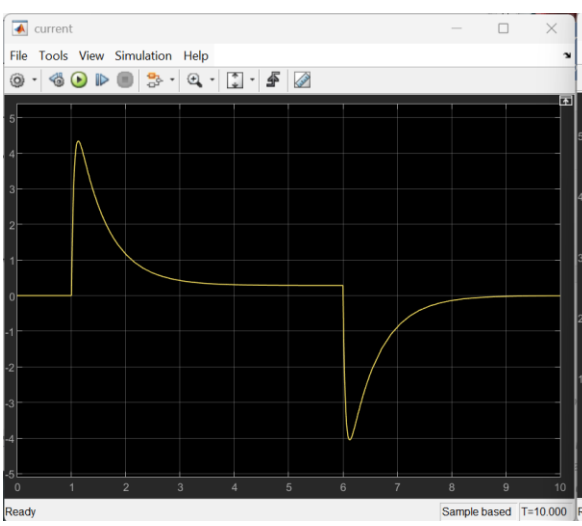
The motor is stationary at time t=1 with maximum current. So, a spike at time t=1 which can be seen in the current simulation. When the motor starts rotating with no load, the system converges to no-load current and the motor speed reaches maximum.

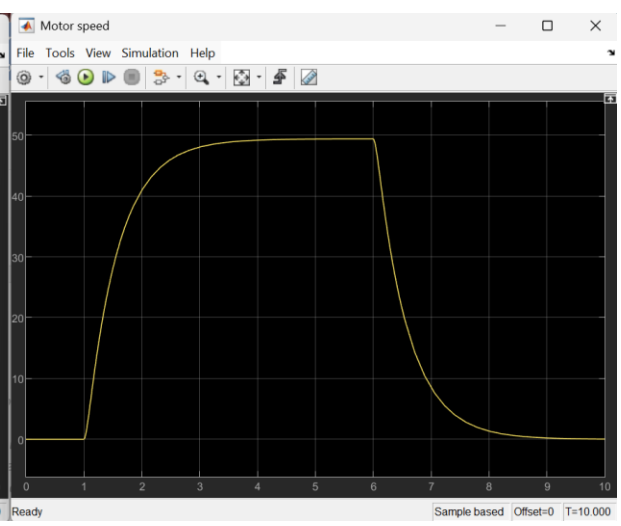
6. Now at the external load-torque input add a step torque of 0.25 Nm (about half of the stall torque) at time t = 6 seconds. Increase the simulation interval to 15 seconds. Explain what you see.





7. Next, keeping the load-torque zero, apply an input voltage which steps from 0V to 12V at t=1 seconds and then drops to 0V again at t=6 seconds. You can achieve this by adding together the signals of two step sources. Explain why you see a negative current spike at t=6 seconds. Hint: Make use of (1.18). The reverse voltage spike which causes this current can damage other components (like transistors) connected to the circuit – this is why, usually a flyback/snubber diode is used.

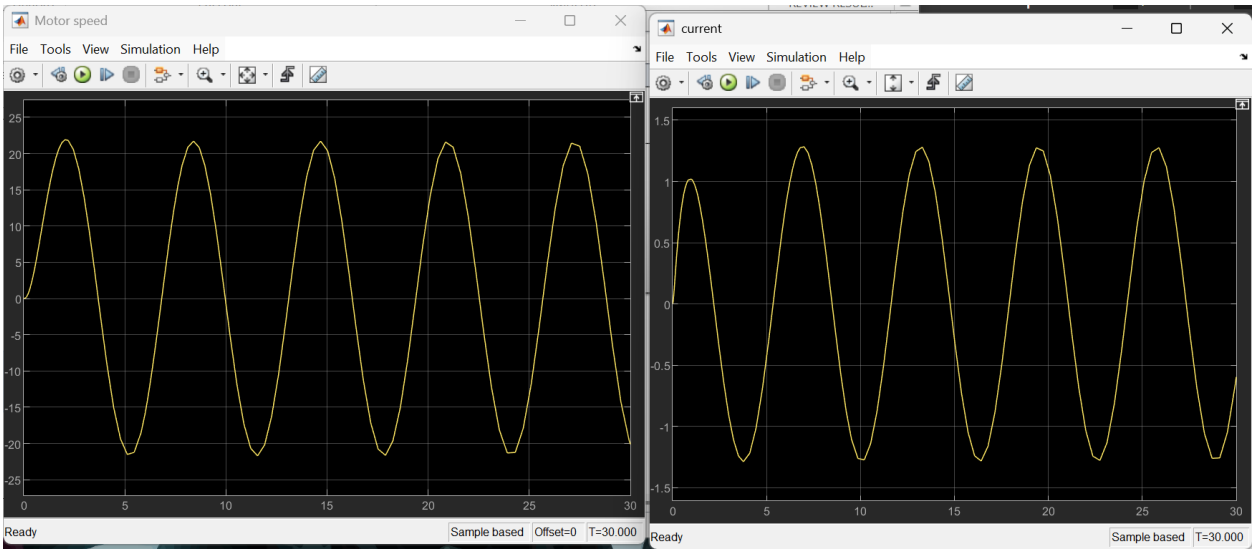




Task 1.11

Use the Source/Sine-Wave block to input a sine-wave voltage of amplitude 6V and frequency 1 rad/s to the motor-model. Simulate for about 30 seconds. After the initial transients have died down and the system reaches steady-state, note down the amplitude I of the sinusoidal current response from the scope – use the zoom-button. Compute the ratio

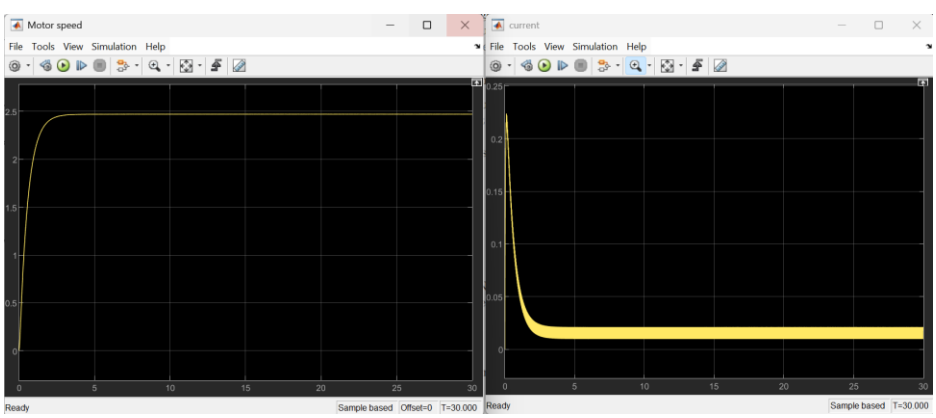
$$h(\omega = 1) = \frac{I}{V_a} = \frac{I}{6}$$



$$\mathrm{h}(\omega = 1) = \frac{\mathrm{I}}{\mathrm{V_a}} = \frac{1.280}{6} = 0.2133$$

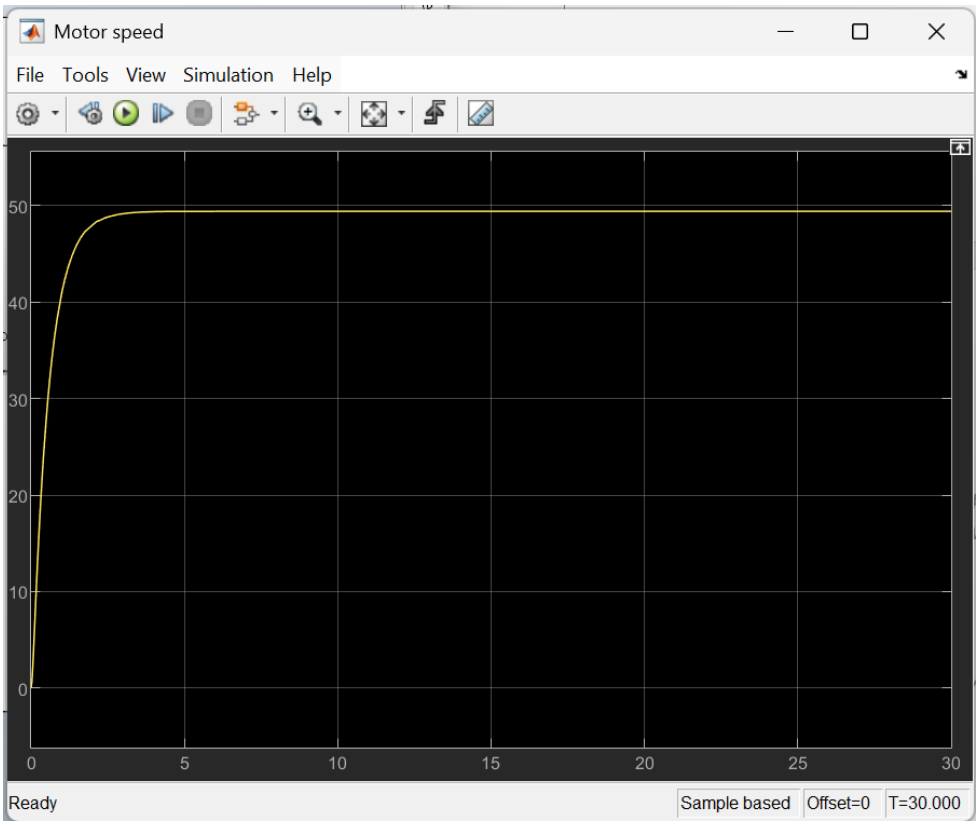
Task 1.12

Set the time-period of the pulse-generator to correspond to that of an Arduino PWM signal, namely 2 milli-seconds. Set the amplitude to be 12 V. Vary the pulse width parameter to vary the PWM duty-cycle value in percentage. See how the motor's speed and current react to such an input.



Task 1.13

In your model, set the load-torque input to zero and apply a step voltage input of 12V at t = 0. Measure Tc from the speed-output scope. Now estimate J. Does it match with the value you knew beforehand?



Zoom in to find the Tc is around 0.635 (since 51.969 x 63% = 32.74047

$$J = \frac{T_c(R_a b + k_t k_e)}{R_a}$$

$$= \frac{0.635\,(2.4 \times 6.798 \times 10^{-4} + 0.11864 \times 0.22918)}{2.4}$$

$$= 7.62567 \times 10^{-3}$$

Therefore, it matches the value of 0.007

Task 1.14.

Find the transfer function HI (s) = I(s)/Va(s) from (1.24) by assuming no-load and eliminating Ω(s) between the two equations.

$$I(s) = \frac{V_a(s)}{L_a s + R_a} - \frac{k_e}{L_a s + R_a} \cdot \frac{k_t}{J_s + b} I(s)$$

$$\frac{V_a(s)}{L_a s + R_a} = I(s)\left[1 + \frac{k_e k_t}{(L_a s + R_a)(J_s + b)}\right]$$

$$\frac{I(s)}{V_a(s)} = \frac{1}{L_a s + R_a}\left[1 + \frac{k_e k_t}{(L_a s + R_a)(J_s + b)}\right]^{-1}$$

$$H_s(s) = \frac{1}{L_a s + R_a}\left[\frac{(L_a s + R_a)(J_s + b) + k_e k_t}{(L_a s + R_a)(J_s + b)}\right]^{-1}$$

$$= \frac{\cancel{(L_a s + R_a)}(J_s + b)}{\cancel{(L_a s + R_a)}\left[(L_a s + R_a)(J_s + b) + k_e k_t\right]}$$

$$= \frac{J_s + b}{(L_a s + R_a)(J_s + b) + k_e k_t}$$

Task 1.15.

We would like to compute |HI (jω)| at ω = 1 using the known parameter values in Matlab. HI (jω) is the complex-number which you get by substituting s = jω in HI (s). Compare |HI (jω)|ω=1 with the ratio h(ω = 1) you found back in (1.25). Are they the same?

```
>> (J*j+b)/((L*j+R)*(J*j+b)+ke*kt)

ans =

   0.1233 + 0.1710i

>> abs(ans)

ans =

    0.2108
```

Task 1.16.

(Bonus) How can you use the result above to find La if it is the only unknown parameter? It will involve solving a fairly nonlinear equation for La, so just give a sketch of the procedure. In literature you also find another method for finding La using the stall-response. However, as this could possibly damage the motor, that method will not be discussed here.

$$H_I(s) = \frac{Js + b}{(L_a s + R_a)(Js + b) + k_e \cdot k_t}$$

$$H_I(s)(L_a s + R_a)(Js + b) + H_I(s) k_e k_t = Js + b$$

$$H_I(s)(L_a s + R_a)(Js + b) = Js + b - H_I(s) k_e k_t$$

$$L_a s = \frac{Js + b - H_I(s) k_e k_t}{H_I(s)(Js + b)} - R_a$$

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